MIDAS GUI — Development History

A commit-by-commit record of what changed and when, so you can (a) find **when** a particular behaviour/feature entered the code and (b) know **what to roll back** to undo a specific effect.

> Keep this file updated when you add commits: append a new entry at the bottom of the chronological log and add a row to the quick-reference index. Regenerate the PDF the same way as the other docs (pandoc + headless Chrome) if you keep one.

How to use this document

- **Find when a change landed:** search the change → commit index below, or `git log --oneline -- <path>` for a specific file, or `git log -S"<code string>"` to find the commit that introduced/removed a string.
- **See a commit's exact diff:** `git show <hash>` (whole commit) or `git show <hash> -- <path>` (one file).
- **Undo one commit cleanly (keeps later history):** `git revert <hash>`. For a range: `git revert <oldest>^..<newest>`.
- **Undo just one file back to a commit:** `git checkout <hash> -- <path>` then commit. To see the file *as of* a commit without changing anything: `git show <hash>:<path>`.
- **Hard reset to a point (destructive — local only, never on pushed history):** `git reset --hard <hash>`. Prefer `git revert` on a shared branch.
- **Dependencies matter.** Many later commits build on earlier ones (e.g. the unified data-loader, the config overlay). The "Roll back" note on each entry flags when reverting will disturb dependents.

Branch: `main`. All commits are authored by Dishant Beniwal (AI pair-authored). Dates are commit dates (YYYY-MM-DD).

Change → commit quick-reference index

Packaging / build / launch

Change	Commit
Initial v1.0.0 release (9-tab app, all backends)	`f19ee8c`
setuptools.build_meta backend (broader pip compat)	`3cd1f36`
`launch.py` standalone launcher	`2e9f419`
Startup crash diagnostics + robust launcher (Windows)	`916ece2`
MIDAS-style packaging (LICENSE, pyproject, release.sh, smoke tests)	`e862135`

Change	Commit
Ship test_data sample data in the repo (fresh clone works out of the box)	`7e157e0`
Drop midas_suite from env (unblock conda env create; backends via -e .)	`72a1770`

Data Viewer (Tab 0)

Change	Commit
Radial-integration plot, beam-centre ring picking, zoom persistence	`acf0866`
Zoom/pan preserved across frames (autoRange off)	`d9e527e`
Intensity-range mask, calibration-file loading, click-to-draw ring	`d50b588`
Compact 3-per-row lattice + cubic quick-set; mask default max(99.99pct,1e5)	`41f28f5`
Projection Skip-frames field; compact geometry/intensity fields; Lsd separators	`108ea7d`
Intensity-statistics panel + histogram (bottom of loader)	`2b7a695`

Mask Builder (Tab 1)

Change	Commit
Unbounded σ fields; independent spatial/temporal auto-mask; Viewer↔Calibrate geometry hand-off	`10df828`
Temporal-constancy accepts a 3-D HDF5 (time,y,x) stack	`2385f75`

Calibrate (Tab 2) / Refinement (Tab 3)

Change	Commit
Abort buttons; dark/bright/background field correction; auto-load on browse	`7d010d7`
Calibration input accepts paramstest/poni/json	`41f28f5`
Load-calibration-file (geometry + λ) helpers	`b381d8d`
Multi-column paramstest results readout + Send-to-Data-Viewer button	`455ca4e`

Batch Integrate (Tab 4)

Change	Commit
Live folder MONITOR + per-file legend	`1149afc`
Clear results + inline per-file labels & plot controls	`524f5f1`
Stacked profiles: publication themes, point+line, symbol size, x-units, grid	`d135c80`
Waterfall R/2θ/Q x-axis selector	`2385f75`

PDF Analysis (Tab 6)

Change	Commit
Polyatomic midas_pdf reduction pipeline + vendored midas_pdf + calibration loading	`b381d8d`
Defaults to real I(Q) test data	`1149afc`

Pump Probe (TR-XRD) tab

Change	Commit
New tab: TRR pooling + MIDAS-engine integration + ΔI(q,delay) + 4 pyqtgraph views + publication-quality plots + default MIDAS calibration	`590b410`

Cross-cutting UI / infrastructure

Change	Commit
Linux visible files/folders style fix	`fc754d0`
Pixel-weighted azimuthal mean; readable QMenu; no-scroll combos	`41f28f5`
Unified Data-Loader panel + draggable 3-panel splitter (tabs 0/2/3/4)	`aa9395e`
Clickable λ label with K-edge foil menu	`b628446`
Clickable px label with detector pixel-size menu	`3248638`
Per-user JSON configuration system + Preferences dialog	`d30be2c`
Modular tab visibility (show/hide optional tabs)	`590b410`
User-adjustable interface scaling (HiDPI / 4K, QT_SCALE_FACTOR)	`d5fafd8`
Default optional tabs reduced (Corrections/PDF/Texture/Export hidden)	`7e157e0`
Lsd displayed in mm (calculations & calibration files stay in μm)	`c4e1c12`
Fix "Populating font family aliases" warning (real fixed-width fonts)	`d6df1f8`
Fix fresh-env launch crash — colormap resolution without matplotlib	`6332b3d`

Stability / performance / consistency (review-driven, phases 1–3)

Change	Commit
Clean worker shutdown, drop QThread.terminate(), guard stdout, offload projection	`399f3d7`
Waterfall $O(N^2) \rightarrow O(N)$, frame-slider debounce, radial-grid + stats caching	`4462ee1`
Dedupe distortion map + paramstest writer, align calibrant lattices, texture colormap	`8846619`

Documentation

Change	Commit
README not-on-PyPI clone/conda instructions	`cd05ced`
gui_documentation.md added	`2e5f7c9`
gui_documentation.pdf added	`75c135c`
README/environment install via `pip install midas_suite`	`aceb40f`
implementation_details.md/.pdf added	`524f5f1`
config_gui.md + config.example.json added	`d30be2c`
development_history.md/.pdf added	`6d6f3fb`

Chronological commit log (oldest → newest)

`f19ee8c` — Initial release: midas_gui v1.0.0 (2026-06-30)

Effect: First version. Nine-tab PyQt5 GUI over `midas_calibrate_v2` / `midas_integrate_v2`: mask building, auto-calibration, refinement, batch integration + waterfall, corrections, PDF, texture, export. Entry points `midas-gui` / `python -m midas_gui`. **Files:** entire `midas_gui/` package + `pyproject.toml`, `README`, `environment.yml`. **Roll back:** this is the root — everything depends on it; don't revert.

`3cd1f36` — setuptools.build_meta backend (2026-06-30)

Effect: Build backend switched for setuptools≥61 compatibility. **Files:** `pyproject.toml`. **Roll back:** `git revert 3cd1f36` (only affects builds).

`2e9f419` — add launch.py standalone launcher (2026-06-30)

Effect: `python /path/to/launch.py` runs the app from anywhere. **Files:** `launch.py`. **Roll back:** safe to revert; later hardened by `916ece2`.

`fc754d0` — Linux style fix for invisible files/folders (2026-06-30)

Effect: Stylesheet fix so file/folder text is visible on Linux. **Files:** `midas_gui/style.py`. **Roll back:** `git revert fc754d0` (cosmetic).

`acf0866` — Data Viewer: radial plot, ring picking, zoom persistence (2026-06-30)

Effect: Added the radial-integration plot, beam-centre ring picking, and zoom-persistence across a stack in the Data Viewer; widened left panels. **Files:** `tab_view.py` (+1-line width bumps in other tabs). **Roll back:** revert to remove the Data Viewer radial plot/ring picking.

`d9e527e` — preserve zoom/pan across frames (2026-06-30)

Effect: Disabled pyqtgraph autoRange on redisplay so zoom/pan survive frame changes. **Files:** `widgets.py`. **Roll back:** revert if you *want* auto-refit per frame.

`d50b588` — Data Viewer: intensity mask, calib loading, click-to-draw ring (2026-07-01)

Effect: 99.9-pct intensity-range mask default; load calibration (json/poni/ paramstest); click-to-draw ring; test-data defaults point at local `test_data/`. **Files:** `tab_view.py`, `widgets.py`, `helpers.py`, `constants.py`, `tab_mask.py`. **Roll back:** revert to drop these Data Viewer features (note `helpers.py` geometry parsing added here is reused later — see `b381d8d`).

`916ece2` — startup crash diagnostics + robust launcher (2026-07-01)

Effect: Logs uncaught exceptions/native faults to `~/midas_gui_error.log`, prevents PyQt slot-exception aborts, isolates failing tabs, hardens the launcher. **Files:** `app.py`, `launch.py`. **Roll back:** revert only if you want raw exceptions to propagate; not recommended.

`7d010d7` — Abort buttons + dark/bright/background + auto-load (2026-07-01)

Effect: Abort on calibrate/batch; dark/bright/background field correction (file/folder/HDF5, index-range averaging, divide/subtract); browse auto-loads, "Load" buttons removed; compact FieldSelector. **Files:** `tab_batch/calibrate/corrections/mask/pdf/refine/texture/view.py`, `widgets.py`, `workers.py`, `helpers.py`. **Roll back:** wide-reaching; reverting removes field correction + auto-load across tabs. The Abort behaviour here was later re-worked by `399f3d7` (no `terminate()`).

`e862135` — MIDAS-style packaging (2026-07-01)

Effect: BSD-3 LICENSE, MIDAS-convention `pyproject.toml`, `release.sh` + `RELEASING.md`, headless `tests/` smoke suite. **Files:** `LICENSE`, `README.md`, `RELEASING.md`, `pyproject.toml`, `release.sh`, `tests/`. **Roll back:** revert affects packaging/tests only.

`41f28f5` — compact lattice, pixel-weighted azimuthal mean, readable menus (2026-07-06)

Effect: Data Viewer 3-per-row lattice + cubic quick-set; intensity-mask default max(99.99pct,1e5); **pixel-weighted azimuthal mean** (selectable vs legacy η-bin mean) fixing off-detector-BC distortion; batch calibration accepts paramstest/poni/ json; readable right-click menus; no-scroll combos. **Files:** `helpers.py`, `workers.py`, `tab_view/batch/calibrate/pdf/refine/texture.py`, `widgets.py`, `style.py`. **Roll back:** to undo the azimuthal-averaging change specifically, prefer editing the `weighted` default rather than reverting the whole commit (it bundles UI changes).

`cd05ced` — docs: not-on-PyPI install instructions (2026-07-06)

Effect: README clone+conda+run instructions; relative env self-install. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

`2e5f7c9` — docs: add gui_documentation.md (2026-07-06)

Effect: First committed user guide. **Files:** `documentation/gui_documentation.md`. **Roll back:** docs only.

`75c135c` — docs: add gui_documentation.pdf (2026-07-06)

Effect: PDF build of the guide. **Files:** `documentation/gui_documentation.pdf`. **Roll back:** docs only.

`aceb40f` — docs: install via `pip install midas_suite` (2026-07-06)

Effect: Corrected backend-install instructions. **Files:** `README.md`, `environment.yml`. **Roll back:** docs only.

`aa9395e` — Unified Data-Loader panel + 3-panel splitter (tabs 0/2/3/4) (2026-07-07)

Effect: Shared `DataLoaderPanel` (Data/Dark/Bright/Background/Mask, per-mode frame controls) + `MaskSelector`; corrections applied on tabs 0/2/3/4; each of those tabs restructured into a draggable `[Data Loader | Parameters | Display]` splitter. **Files:** `widgets.py` (+442), `tab_view/calibrate/batch/refine.py` (large rewrites), `app.py`, `helpers.py`, `workers.py`, docs. **Roll back: high impact** — this is the base for the Batch/Calibrate/Refine and later Pump Probe layouts. Reverting breaks the 3-panel tabs and the Pump Probe tab's left panel. Avoid a blanket revert; fix forward instead.

`b381d8d` — Tab 6 polyatomic midas_pdf pipeline + calibration loading (2026-07-08)

Effect: PDF tab rewritten to the total-scattering pipeline; **vendored `midas_pdf`** under `midas_gui/_vendor/` + `pdf_backend.py`; `helpers` `geometry_fields_from_file` / `result_ns_from_geometry_file` (paramstest/json/poni) used by Calibrate and PDF "Load calibration file…". **Files:** `_vendor/midas_pdf/**` (new, large), `pdf_backend.py`, `tab_pdf.py`, `workers.py`, `helpers.py`, `tab_calibrate.py`, `constants.py`, `pyproject.toml`, docs. **Roll back:** reverting removes the PDF pipeline **and** the shared geometry-file loaders that later tabs (incl. Pump Probe via `spec_from_geometry_file`) rely on — revert with care.

`1149afc` — Batch live MONITOR + legend; PDF real-data default (2026-07-08)

Effect: MONITOR button (stream mode) watches a folder and integrates new frames via `FolderMonitorWorker`, reusing a cached detector map (factored `build_integration_context()` + `write_profile()`, shared with `BatchWorker`); per-file legend; PDF tab defaults to real I(Q). **Files:** `tab_batch.py`, `workers.py`, `widgets.py`, `constants.py`, `tab_pdf.py`, docs. **Roll back:** to remove MONITOR only, revert this commit — but `build_integration_context()` introduced here is **reused by the Pump Probe worker** (`590b410`), so a full revert would break Pump Probe integration.

`524f5f1` — Batch Clear results + inline labels; implementation-details doc (2026-07-09)

Effect: "Clear results" button; inline per-file curve labels + line-width/font toolbar on stacked profiles; new `implementation_details.md/.pdf`. **Files:** `tab_batch.py`, `widgets.py`, `documentation/implementation_details.*`. **Roll back:** safe, localized to Batch + the new doc.

`10df828` — Mask unbounded σ + split auto-mask; Viewer↔Calibrate geometry (2026-07-09)

Effect: Removed σ upper limits; split statistical auto-mask into independent Spatial-outlier / Temporal-constancy checkboxes; geometry hand-off `DataViewer.pushGeometry` / `Calibrate.pullGeometry` → `apply_geometry`. **Files:** `tab_mask.py`, `tab_view.py`, `tab_calibrate.py`, `app.py`, `workers.py`, docs. **Roll back:** revert to restore bounded σ / combined auto-mask / no geometry hand-off.

`d135c80` — Batch stacked-profiles: themes, point+line, x-units, grid (2026-07-09)

Effect: `StackedProfileViewer` publication/dark themes, point+line, symbol-size, R/2θ/Q x-unit selector, grid toggle. (This viewer's styling is the template the Pump Probe plots later mirror.) **Files:** `widgets.py` (+228), `tab_batch.py`, docs. **Roll back:** localized to the Batch stacked-profile viewer.

`2385f75` — Batch waterfall R/2θ/Q axis; Mask 3-D HDF5 stack (2026-07-10)

Effect: Waterfall x-unit selector via a converting `AxisItem` + `_convert_radial()`; Mask temporal-constancy reads a single 3-D (time,y,x) HDF5. **Files:** `widgets.py`, `tab_batch.py`, `tab_mask.py`, `workers.py`, docs. **Roll back:** `_convert_radial` / `_UnitAxis` introduced here are reused by the Pump Probe heatmap — don't fully revert if Pump Probe is present.

`b628446` — clickable λ label with K-edge foil menu (2026-07-10)

Effect: Compact clickable "λ" label → K-edge foils menu (sets λ from edge energy). `constants.K_EDGE_FOILS`, `HC_KEV_A`, `helpers.make_kedge_label`. **Files:** `constants.py`, `helpers.py`, `tab_view/calibrate/pdf.py`, docs. **Roll back:** localized; `make_pixel_label` (next) reuses the factored helper.

`108ea7d` — Data Viewer projection Skip-frames + compact fields (2026-07-10)

Effect: Projection "Skip frames" field; narrower Axis/Skip/pixel fields; Lsd thousands-separators. **Files:** `tab_view.py`, docs. **Roll back:** localized.

`3248638` — clickable px label with detector pixel-size menu (2026-07-10)

Effect: Clickable "px" label → GE/Varex/Pilatus/Eiger sizes; `constants.PIXEL_PRESETS`; factored `helpers._clickable_menu_label`. **Files:** `constants.py`, `helpers.py`, `tab_view.py`, `tab_calibrate.py`, docs. **Roll back:** localized.

`2b7a695` — Data Viewer intensity-statistics panel + histogram (2026-07-10)

Effect: `IntensityStatsPanel` (histogram + p70/p90/p99/p99.9/p99.99 with pixel counts) pinned to the loader bottom (stack mode); Current/All/projected scope. **Files:** `widgets.py`, `tab_view.py`, docs. **Roll back:** localized.

`d30be2c` — per-user configuration system + Preferences dialog (2026-07-12)

Effect: Per-user JSON config overlays shipped defaults at import; `settings.py`; `prefs_dialog.py` (Geometry/Paths/Materials/Calibrants/Menus/Algorithms); Settings menu; `DEFAULT_KERNEL/PIPELINE/OUTPUT_FORMAT/ERROR_MODEL/COLORMAP`; tab combos honor configured defaults. New `config_gui.md`, `config.example.json`, `tests/test_config.py`. **Files:** `settings.py` (new), `prefs_dialog.py` (new), `constants.py` (+165), `app.py`, `tab_batch.py`, `tab_calibrate.py`, `widgets.py`, docs, tests. **Roll back: high impact** — the config overlay + `DEFAULT_*` and the Preferences dialog are extended by the modular-tabs commit (`590b410`). Reverting this would also break `ui.visible_tabs` and the Tabs preferences section.

`399f3d7` — Stability: clean worker shutdown, no terminate(), stdout guard (2026-07-12)

Effect: `closeEvent` stops all tab QThreads; removed every `QThread.terminate()` (cooperative interrupt + orphan instead); `CalibrationWorker` restores stdout only if still its own; Data Viewer projection moved to `ProjectionWorker` (off GUI thread). **Files:** `app.py`, `tab_batch.py`, `tab_calibrate.py`, `tab_view.py`, `workers.py`. **Roll back:** reverting reintroduces the exit hard-crash risk and UI-freeze on projection — not recommended. Phase 1 of the 3-part review.

`4462ee1` — Performance: waterfall O(N), debounce, caching (2026-07-12)

Effect: Waterfall pre-allocated buffer + 100 ms coalesced redraw (O(N) not $O(N^2)$); 60 ms frame-slider debounce; cached radial bin-index grid; skip all-frame stats on frame-only changes. **Files:** `tab_view.py`, `widgets.py`. **Roll back:** reverting slows large scans / scrubbing; behaviour otherwise same. Phase 2 of the review.

`8846619` — Consistency: dedupe maps/writer, align lattices (2026-07-12)

Effect: `helpers._PARAMSTEST_DISTORTION` derived from `constants._V2_TO_V1` (removed duplicated `_V2V1` dicts); shared `helpers.write_standalone_paramstest` used by Calibrate + Export; LaB6/Si `_LATT/_LC` aligned to `MATERIALS`; texture colormap honors `DEFAULT_COLORMAP`. **Files:** `constants.py`, `helpers.py`, `tab_calibrate.py`, `tab_export.py`, `tab_texture.py`. **Roll back:** reverting reintroduces duplicated code paths; low functional risk. Phase 3 of the review.

`590b410` — Modular tabs + Pump Probe (TR-XRD) tab + plot quality (2026-07-12)

Effect: (1) **Modular tab visibility** — Data Viewer/Mask/Calibrate/Batch always shown, the rest toggled in Settings ▸ Preferences ▸ Tabs (`ui.visible_tabs`, applied live); `app.apply_tab_visibility()`. (2) **New Pump Probe tab** (`tab_pumpprobe.py`) — TRR filename pooling, MIDAS-engine integration via new `PumpProbeWorker` (reuses `build_integration_context`/`integrate_frame`), ΔI(q,delay), three-panel layout, four pyqtgraph views; ships a converted MIDAS `paramstest` as the default calibration. (3) **Plot quality** — white publication theme, black axes/labels/titles, readable legends, ΔI colorbar, aligned reference panel, shared draw-mode + line/point/font-size toolbar, bounded pan/zoom. **Files:** `tab_pumpprobe.py` (new), `workers.py` (+`PumpProbeWorker`), `app.py`, `constants.py`, `prefs_dialog.py`, `tests/test_smoke.py`, docs (`gui_documentation`, `config_gui`, `config.example.json`). **Roll back:** to remove **only Pump Probe**, delete `tab_pumpprobe.py` and its wiring in `app.py`/`workers.py`; to remove **only modular tabs**, revert the `app.apply_tab_visibility` / `prefs_dialog` Tabs section and `constants` tab lists. A full `git revert 590b410` removes both features together. Depends on `aa9395e` (DataLoaderPanel), `1149afc` (build_integration_context), `2385f75` (`_convert_radial`/`_UnitAxis`), `d30be2c` (config/prefs).

`6d6f3fb` — docs: add development_history.md/.pdf (2026-07-13)

Effect: Added this per-commit change log + change→commit index + rollback guide. **Files:** `documentation/development_history.md`, `documentation/development_history.pdf`. **Roll back:** docs only.

`d5fafd8` — UI scaling: user-adjustable whole-interface zoom (HiDPI / 4K) (2026-07-13)

Effect: Whole-application scale (layout + fonts) for HiDPI / 4K monitors. `constants.DEFAULT_UI_SCALE` (config `ui.ui_scale`); `app.main()` applies it via `QT_SCALE_FACTOR` **before** the QApplication is created (so fixed widths, splitters, pyqtgraph and stylesheet px all scale uniformly) + `AA_UseHighDpiPixmaps`. Manual control in **Preferences ▸ Display** (spin + 100/125/150/200 % presets) and **Settings ▸ Interface scaling…**; both persist `ui.ui_scale` and offer an immediate self-restart (`MainWindow.restart_app` via `QProcess`) since the factor is read only at startup. **Files:** `constants.py`, `app.py`, `prefs_dialog.py`, docs (`gui_documentation`, `config_gui`, `config.example.json`). **Roll back:** `git revert d5fafd8`. Self-contained (depends only on the config system `d30be2c`); reverting removes the scale control and the app renders at 1.0×. To keep the feature but disable it, set `ui.ui_scale` to 1.0.

`c4e1c12` — Display Lsd in mm (calculations & files stay in μm) (2026-07-13)

Effect: The sample-to-detector distance is entered/shown in **mm** in the Data Viewer, the Calibrate seed, and Preferences ▸ Geometry; conversion to/from μm happens only at the display boundary (`DataViewerTab._lsd_um()`, ×1000 on read, ÷1000 on set). `get_geometry`/`apply_geometry`/`manual_seed`/prefs `_assemble` still emit μm; the config key stays `lsd_um` (μm); calibration-file writers (`paramstest.txt`, `calibration.json`) are unchanged (always μm). Batch drift-status label → mm. **Files:** `tab_view.py`, `tab_calibrate.py`, `prefs_dialog.py`, `tab_batch.py`, docs. **Roll back:** `git revert c4e1c12` to return every Lsd field to a μm display. Purely a display-layer change — no stored/file/calculation values are affected either way.

`d6df1f8` — Fix "Populating font family aliases" startup warning (2026-07-13)

Effect: Removed the `qt.qpa.fonts: Populating font family aliases` warning (and its ~40 ms startup cost) that Qt emitted because the code named the nonexistent family "Monospace" — both via `QFont("Monospace", …)` and the CSS generic `font-family:monospace`. Now names real per-platform families (Menlo/Consolas/DejaVu Sans Mono/Courier New) via `style.MONO_FAMILIES` / `MONO_CSS` and `widgets._mono_font()` (QFont `setFamilies` + Monospace style hint). **Files:** `style.py`, `widgets.py`, `tab_export.py`, `tab_view.py`. **Roll back:** `git revert d6df1f8` (cosmetic; only affects fixed-width font selection and re-introduces the harmless warning).

`455ca4e` — Calibrate: multi-column results readout + Send-to-Data-Viewer (2026-07-13)

Effect: (1) The Calibrate **Results** tab shows the full parameter set exactly as written to `paramstest.txt` (Lsd, BC, tx/ty/tz, p0–p14, Parallax, Wavelength, px, NrPixelsY/Z, RhoD, SpaceGroup, LatticeConstant) in a 3-column, column-major grid (`_paramstest_pairs` via the shared writer + `_populate_param_grid`), with a strain/timing line and the named-distortion table below. (2) New → **Send to Data Viewer** button (`sendGeometryToViewer` signal + `_send_to_viewer`) pushes the calibrated λ/pixel/Lsd/beam-centre into the Data Viewer through new `DataViewerTab.set_geometry` (μm internal, Lsd shown mm, auto-BC off); wired in `app.py` as the reverse of the Viewer→Calibrate hand-off. **Files:** `tab_calibrate.py`, `tab_view.py`, `app.py`, docs. **Roll back:** `git revert 455ca4e`. Self-contained; depends on the mm-display change (`c4e1c12`) for the Viewer Lsd conversion.

`7e157e0` — Ship test_data; hide four optional tabs by default (2026-07-14)

Effect: (1) `test_data/` sample data (calibrant_ceria.tif/h5, nickel_stack.h5, nickel_tifs/, make_test_data.py) is now committed so the GUI's default paths work on a fresh clone — `.gitignore` re-includes it past the global `*.tif` / `*.h5` ignores via trailing negations, while `test_data/output/`, `out_temp.txt` and `.DS_Store` stay ignored. (2) `DEFAULT_VISIBLE_TABS = ["Calib. Refinement", "Pump Probe"]` — Corrections, PDF Analysis, Texture and Results & Export ship hidden (enable in Preferences ▸ Tabs). **Files:** `.gitignore`, `test_data/**` (added), `constants.py`, `config.example.json`, `tests/test_smoke.py`, docs. **Roll back:** `git revert 7e157e0` restores the all-optional-tabs default and removes the shipped data + re-ignores `test_data/`. Note: a per-user config with its own `ui.visible_tabs` overrides the shipped default regardless.

`6332b3d` — Fix launch crash on fresh Linux/Windows (colormap w/o matplotlib) (2026-07-14)

Effect: Fixes the GUI failing to start on a clean env where every always-on tab's image viewer crashed with `'NoneType' object has no attribute 'getColors'`. Cause: `DEFAULT_COLORMAP` "hot" (and the whole `COLORMAPS` list) are matplotlib colormaps; `pg.colormap.get("hot")` returns None without matplotlib, and that None was passed into pyqtgraph. New `widgets._resolve_cmap()` resolves native → matplotlib → native fallback → grayscale ramp and never returns None (used by ImageViewer, WaterfallViewer, Texture); matplotlib added as a declared dependency (`pyproject` + `environment.yml` matplotlib-base) so the named maps actually resolve. The downstream "Geometry hand-off wiring failed / QWidget has no attribute pushGeometry" log lines were only a symptom of the tabs becoming placeholders and disappear with this fix. **Files:** `widgets.py`, `tab_texture.py`, `pyproject.toml`, `environment.yml`, `tests/test_smoke.py`. **Roll back:** `git revert 6332b3d` (reintroduces the crash on matplotlib-less envs).

`72a1770` — env: drop midas_suite (unblocks `conda env create`) (2026-07-14)

Effect: `conda env create -f environment.yml` was aborting because the pip section's `midas_suite` meta-package pulls `midas-index 0.7.3`, whose sdist fails to build against `scikit-build-core>=0.8` (`Use cmake.version instead of cmake.minimum-version`). The GUI never uses midas_suite's extras; the backends it actually imports (midas-calibrate-v2/-integrate-v2/-calibrate/-hkls/-distortion) are declared in `pyproject.toml` and pulled by the editable `-e .` install. Removed the redundant `midas_suite` line; README updated. (Unrelated to the repo: the `libarchive.19.dylib` / conda-libmamba-solver errors in that log are a broken base-Anaconda mamba, worked around with `--solver=classic`.) **Files:** `environment.yml`, `README.md`. **Roll back:** `git revert 72a1770` (re-adds midas_suite and its build failure).

- **Undo the Pump Probe tab only:** `git revert 590b410` removes Pump Probe *and* modular tabs. To drop Pump Probe alone, hand-remove `midas_gui/tab_pumpprobe.py`, its import/construction/wiring in `app.py`, `PumpProbeWorker` in `workers.py`, and its entry in `constants.OPTIONAL_TABS` / `DEFAULT_VISIBLE_TABS`.
- **Undo modular tabs (show all 10 fixed again):** remove the `apply_tab_visibility` logic in `app.py` (add all tabs directly), the Tabs section in `prefs_dialog.py`, and the tab-list constants — see the `590b410` diff for the exact hunks.
- **Undo the config system:** `git revert d30be2c` — but first revert `590b410`'s config touches, since it extends the same `ui` overlay and Preferences dialog.

- **Revert the azimuthal-averaging change:** don't blanket-revert `41f28f5` (bundled with UI); instead set the `weighted` default back to the legacy η-bin mean in the batch/pump workers.
- **Go back to the pre-3-panel single-column tabs:** `aa9395e` is the pivot; a revert is large and cascades to later tabs — prefer fixing forward.

Generated from `git log`. To refresh after new commits: `git log --reverse --stat --date=short` and append entries above.